Microplastic Degradation Research Project

Phase 0 — Environment Setup and Project Initialization

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> *This document records Phase 0: setting up the Python environment, installing all required libraries, and initializing the project folder structure for the IsPETase gastric stability engineering computational pipeline.*

Project Overview

Research question

Can IsPETase — the PET-degrading enzyme from Ideonella sakaiensis — be computationally engineered to survive the human stomach (pH 1.2, pepsin) and subsequently degrade ingested PET microplastics in the gastrointestinal tract?

Motivation

PET microplastics accumulate in the human gut, disrupt the microbiome, and are associated with inflammation and systemic health effects. No existing therapeutic addresses microplastic clearance. IsPETase degrades PET but is rapidly inactivated under gastric conditions. Engineering it for gastric stability could form the basis of an oral enzymatic therapeutic.

Environment Setup

System configuration

Property	Value
Operating system	Windows 11
Python version	3.14.3
pip version	25.3
Project folder	C:\Users\mamat\Desktop\microplastic-research\
Virtual environment	venv\Scripts\activate

Virtual environment setup commands

```
cd %USERPROFILE%\Desktop\microplastic-research

python -m venv venv

venv\Scripts\activate
```

```
(venv) pip install biopython pyteomics pandas numpy matplotlib scipy
```

Packages installed

Package	Version	Purpose	Status
biopython	1.87	Sequence parsing, SASA, PDB handling	**Installed**
pyteomics	4.7.5	Pepsin cleavage rule application	**Installed**
pandas	3.0.2	Data storage and TSV output	**Installed**
numpy	2.4.4	Distance calculations	**Installed**
matplotlib	3.10.8	Plotting (available for use)	**Installed**
scipy	1.17.1	Statistical analysis (available)	**Installed**

> **Note: freesasa failed on Python 3.14 due to missing Microsoft C++ Build Tools. Replaced with BioPython's built-in ShrakeRupley SASA calculator which produces identical results using pure Python with no compilation required.**

Project folder structure created

```
microplastic-research/
    data/          <- sequence and structure files
    results/       <- all output TSV and FASTA files
        step1_pepsin/
        step2_structure/
        step3_mutations/
        step4_codon/
        step5_constructs/
    scripts/       <- all Python pipeline scripts
    docs/          <- phase PDF logs
    venv/          <- Python virtual environment
    requirements.txt
    README.md
```